Finding High-Need Countries through Data Mining

Course: Data Mining and Knowledge Discovery Sessional

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Abstract

This project addresses the challenge of identifying countries in critical need of humanitarian aid by analyzing socio-economic and health-related factors. Using a dataset of 167 countries characterized by ten key features—including **child mortality**, **income**, and **GDP per capita**—we employed data mining techniques to categorize nations based on their development status. Our methodology involved rigorous **data cleaning**,**Z-score standardization**, and **exploratory data analysis**, followed by a comparative study between **K-Means** and **hierarchical** clustering algorithms. Evaluation metrics, including *Silhouette Score* , *Davies-Bouldin Index*, and *Calinski-Harabasz Index*, demonstrated that K-Means clustering with four clusters provided a more distinct and robust segmentation than hierarchical clustering. Specifically, the model successfully identified a **high-need** cluster characterized by significantly high child mortality rates and low per capita income. These results provide a data-driven framework for organizations like *HELP International* to prioritize resource allocation effectively.

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1 Introduction

1.1 Background of the Problem

HELP International is an international humanitarian NGO that is committed to fighting poverty and providing the people of backward countries with basic amenities and relief during the time of disasters and natural calamities. After a successful fundraising campaign, the CEO needs to decide how to allocate $10 million effectively. The challenge lies in identifying the countries that are in the most dire need of aid based on socio-economic and health factors.

1.2 Importance and Relevance

Data-driven decision-making is crucial in humanitarian efforts to ensure resources are not wasted. By applying clustering algorithms, we can categorize countries into stages of development without human bias. This allows organizations to prioritize intervention in regions where child mortality is high and economic resources are scarce.

1.3 Research Questions and Objectives

The primary objective of this project is to categorize countries using socio-economic and health factors to determine which ones should be prioritized for aid. Specifically, the project aims to:

- Perform exploratory data analysis to understand the correlation between wealth and health.
- Apply and compare K-Means and Hierarchical Clustering models.
- Identify the optimal number of clusters using metrics like the Silhouette Score.
- List the specific countries that fall into the "High-Need" category.

1.4 Description of the Dataset

The dataset consists of 167 countries [1], each described by 10 features:

- **Health:** Child Mortality, Health spending, Life Expectancy, and Total Fertility.
- **Economics:** Income per person, Inflation, and GDP per capita.
- **Trade:** Exports and Imports as a percentage of the GDP per capita.

Data preprocessing included Z-score normalization to ensure that features with larger numerical ranges (like Income) did not disproportionately influence the clustering results.

2 Dataset Description

2.1 Source of Dataset

The dataset used in this project is a publicly available country-level socio-economic and health dataset, commonly utilized for development analysis and educational data mining projects. It compiles indicators collected from international organizations such as global development and health agencies. The dataset is appropriate for identifying patterns related to national development and well-being.

2.2 Number of Samples and Features

The dataset contains a total of **167 countries** as samples and **9 attributes** as features. Each record corresponds to one country, and each feature represents a specific socio-economic or health-related indicator used for clustering analysis.

2.3 Types of Variables

The variables in the dataset can be categorized as follows:

- **Numerical variables:** child mortality rate (`child_mort`), life expectancy (`life_expec`), total fertility rate (`total_fer`), inflation rate (`inflation`), health expenditure (`health`), income, and GDP per capita (`gdpp`).
- **Categorical variable:** country name, which is used only for identification and visualization purposes and is excluded from clustering computations.

All numerical variables were normalized to ensure equal contribution during distance-based clustering.

2.4 Domain Context

From a development perspective, higher child mortality, higher fertility rates, lower life expectancy, lower income, and lower GDP per capita indicate poorer socio-economic conditions and greater development needs. In contrast, countries with lower mortality rates, higher income levels, and better health indicators are generally more developed. This domain knowledge is essential for interpreting the resulting clusters as High-Need or Low-Need country groups.

2.5 Sample Data Description

Table 1 presents a few representative rows from the dataset to demonstrate its structure and feature composition.

Table 1: Sample Rows from the Dataset

country	child_mort	exports	health	imports	income	inflation	life_expec	total_fe	gdpp
Afghanistan	90.2	10.0	7.58	44.9	1610	9.440	56.2	5.82	553
Albania	16.6	28.0	6.55	48.6	9930	4.490	76.3	1.65	4090
Algeria	27.3	38.4	4.17	31.4	12900	16.100	76.5	2.89	446
Angola	119.0	62.3	2.85	42.9	5900	22.400	60.1	6.16	3530
Antigua and Barbuda	10.3	45.5	6.03	58.9	19100	1.440	76.8	2.13	12200
Argentina	14.5	18.9	8.10	16.0	18700	20.900	75.8	2.37	10300
Armenia	18.1	20.8	4.40	45.3	6700	7.770	73.3	1.69	3220
Australia	4.8	19.8	8.73	20.9	41400	1.160	82.0	1.93	51900
Austria	4.3	51.3	11.00	47.8	43200	0.873	80.5	1.44	46900
Azerbaijan	39.2	54.3	5.88	20.7	16000	13.800	69.1	1.92	5840
Bahamas	13.8	35.0	7.89	43.7	22900	-0.393	73.8	1.86	28000
Bahrain	8.6	69.5	4.97	50.9	41100	7.440	76.0	2.16	20700
Bangladesh	49.4	16.0	3.52	21.8	2440	7.140	70.4	2.33	758

3 Data Cleaning & Processing

3.1 Assessment of Data Quality

The dataset was evaluated for integrity before any analysis was performed.

- **Missing Values:** A check for null values was conducted across all columns. The result showed that the dataset contains no missing values, requiring no data imputation.
- **Duplicates:** The dataset was screened for duplicate rows to prevent bias in the clustering process. No duplicate entries were identified.
- **Outliers:** Exploratory analysis revealed a wide range of values in features like Income and GDPP. These were identified as natural socio-economic variations rather than data entry errors.

3.2 Data Transformations

Proper scaling is essential for distance-based algorithms like K-Means and Hierarchical Clustering.

- **Feature Separation:** The `country` column was set aside as an identifier, ensuring only numeric socio-economic data was used for model training.
- **Standardization:** We applied Z-score standardization to all numeric features. This process transforms the data such that each feature has a mean of 0 and a standard deviation of 1, preventing features with larger scales (like Income) from dominating the distance calculations.

3.3 Reproducibility Documentation

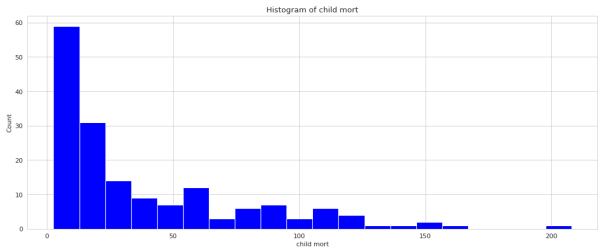
The project maintains a clear workflow within the Python environment. Data was processed using the `pandas` [3] and `numpy` libraries, with visualizations generated via `matplotlib` [5] and `seaborn`[6]. This modular structure ensures that the data cleaning and processing steps can be easily replicated.

4 Exploratory Data Analysis (EDA)

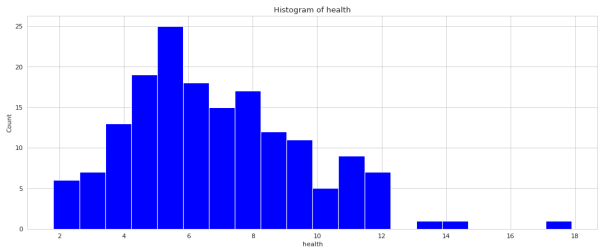
This section presents an exploratory data analysis of the dataset to uncover patterns, relationships, and anomalies across socio-economic and health indicators for different countries. Visualizations are used to understand variable distributions and inter-variable dependencies, providing a foundation for subsequent modeling and clustering.

4.1 Univariate Analysis

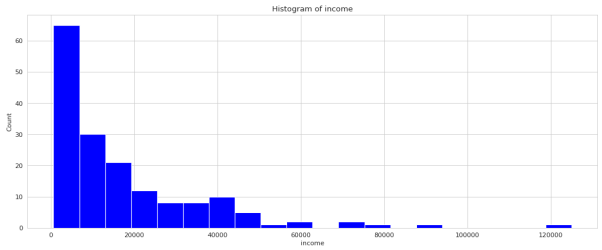
Univariate analysis focuses on understanding the distribution of individual variables. Histograms and box plots were generated for all numerical features to assess central tendency, spread, distributional shape (skewness), and potential outliers. Descriptive statistics (e.g., mean, median, standard deviation, quartiles) were also examined to complement the visual analysis.



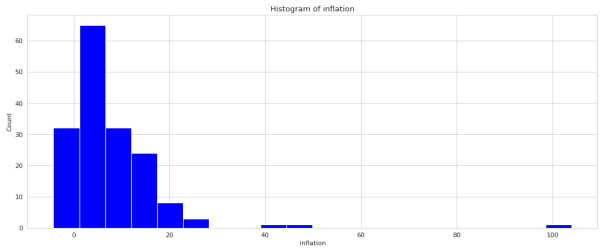
(a) Child mortality



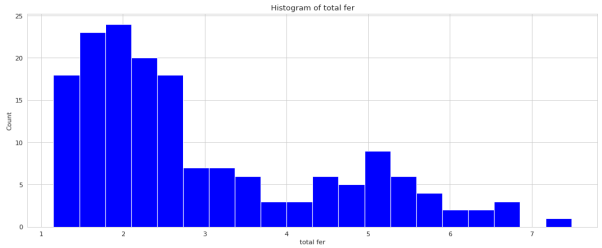
(b) Health spending



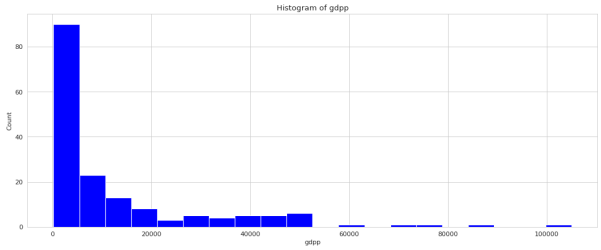
(c) Income



(d) Inflation



(e) Total fertility



(f) GDP per capita (GDPP)

Figure 1: Histograms of selected features.

4.1.1 Histograms

Histograms were created for each numerical variable to visualize frequency distributions and identify skewness, modality, and overall shape. Key observations include:

- **Child Mortality:** The distribution is right-skewed, indicating most countries have relatively low child mortality rates, while a subset experiences very high rates.
- **Exports:** Right-skewed, suggesting many countries have modest export shares, with a few highly export-reliant economies.
- **Health Spending:** Right-skewed, with most countries allocating a smaller fraction of spending to health and fewer allocating substantially more.
- **Imports:** Right-skewed, similar to exports, reflecting that most countries have moderate import shares and a small number have very high import dependence.
- **Income:** Highly right-skewed, showing a large concentration of lower-income countries and a long tail of very high-income countries.
- **Inflation:** Right-skewed, with most countries in low-to-moderate inflation ranges and some extreme high-inflation cases.
- **Life Expectancy:** Roughly symmetric to slightly left-skewed, with many countries clustered at higher life expectancy values and fewer at the lower end.
- **Total Fertility:** Right-skewed, with many countries having lower fertility and some exhibiting very high fertility.
- **GDP per Capita:** Highly right-skewed, reflecting wide economic disparity with many low-GDP-per-capita countries and a small number of very high-GDP-per-capita countries.

Overall, these distributions reflect substantial cross-country disparities in development and health outcomes.

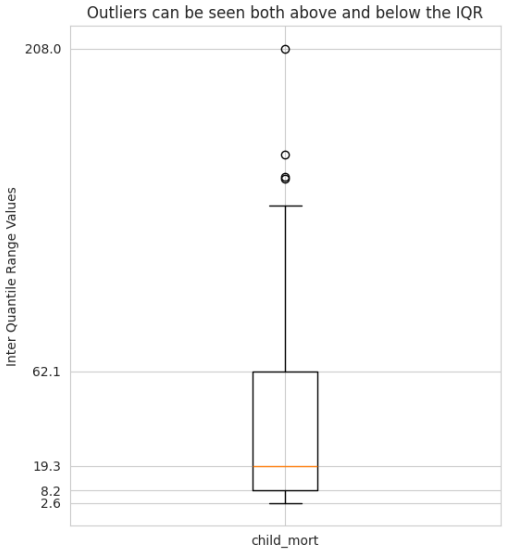
4.1.2 Box Plots

Box plots were used to summarize each numerical variable via the five-number summary (minimum, first quartile, median, third quartile, maximum) and to highlight outliers. Notable findings include:

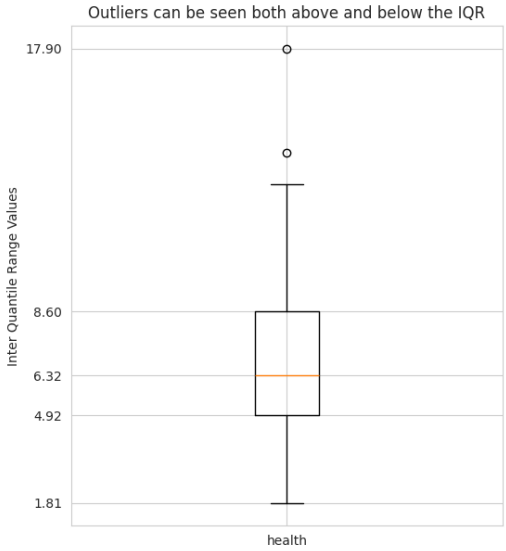
- **Child Mortality:** Many high-end outliers, indicating a set of countries with exceptionally high mortality rates.
- **Exports and Imports:** Upper-end outliers suggest some countries have unusually large trade shares relative to GDP.
- **Health Spending:** High-end outliers point to countries with disproportionately large health expenditure shares.
- **Income and GDP per Capita:** Numerous high-value outliers underline substantial inequality across countries, with a small set far above typical levels.

- **Inflation:** Several extreme outliers indicate episodes of very high inflation in certain countries.
- **Life Expectancy:** Relatively concentrated, though outliers exist at both ends, reflecting exceptionally low or high life expectancy in some countries.
- **Total Fertility:** High-end outliers correspond to countries with very high fertility rates.

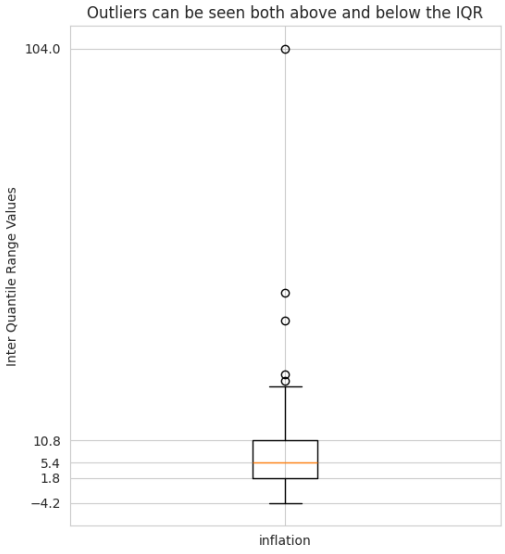
These plots confirm the presence of influential extreme cases, which are important for both interpretation and modeling decisions (e.g., scaling and robust methods).



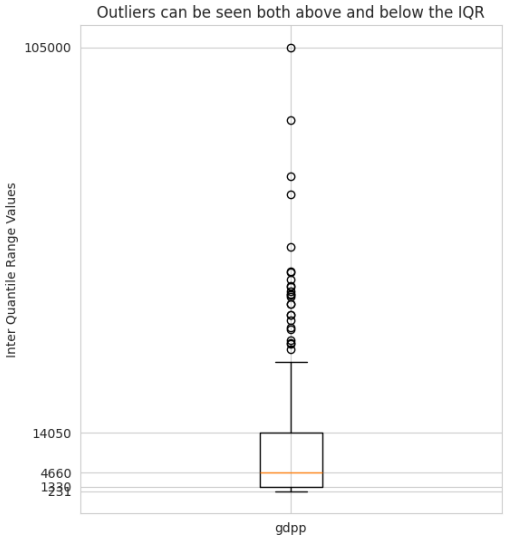
(a) Child mortality



(b) Health spending



(c) Inflation



(d) GDPP

Figure 2: Box Plot of selected features.

4.2 Bivariate and Multivariate Analysis

Bivariate and multivariate analyses were conducted to explore relationships between variables, identify correlated features, and detect potential clustering structures. Scatter plots, pair plots, and a correlation heatmap were used to summarize these dependencies.

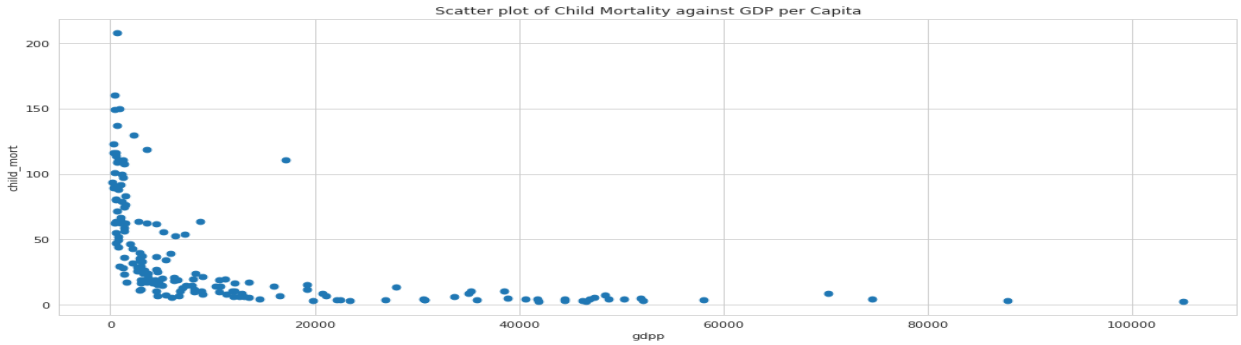


Figure 3: Child Mortality vs. GDP per Capita

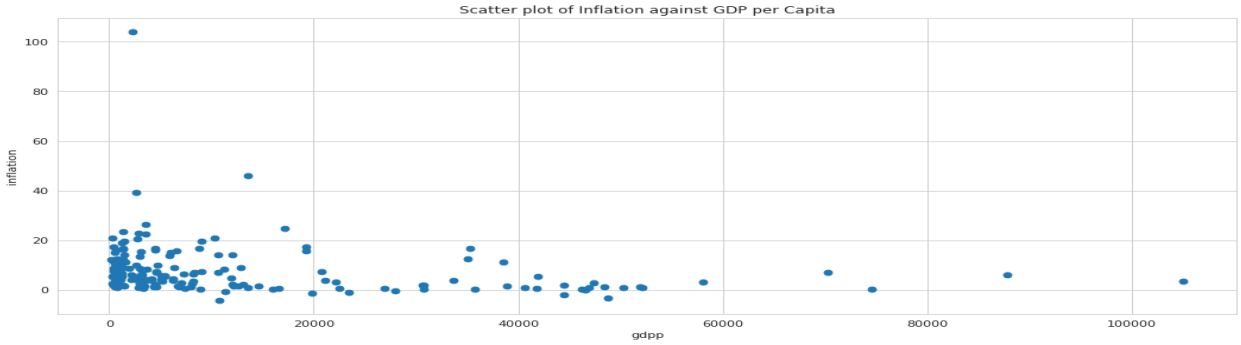


Figure 4: Inflation vs. GDP per Capita

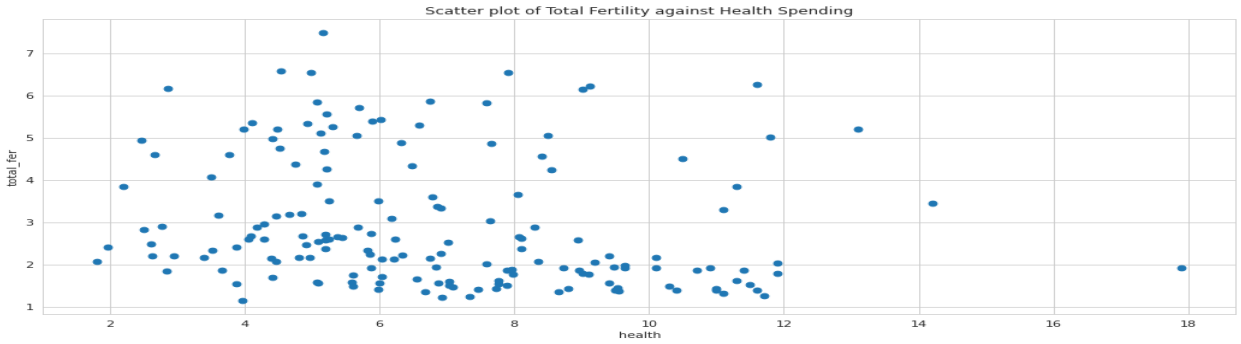


Figure 5: Total Fertility vs. Health Spending

4.2.1 Scatterplots

Scatter plots were generated for selected variable pairs to reveal trends, correlation direction, and potential clusters:

- **Child Mortality vs. GDP per Capita:** A strong negative relationship is observed, where countries with higher GDP per capita generally have lower child mortality, highlighting the link between economic development and child health.
- **Life Expectancy vs. Income:** A clear positive association suggests higher income is linked with higher life expectancy, consistent with improved access to nutrition, sanitation, and healthcare.
- **Total Fertility vs. Health Spending:** A general inverse relationship indicates that higher health investment is often associated with lower fertility, potentially reflecting better healthcare access and education.
- **Inflation vs. GDP per Capita:** No strong linear pattern is evident, though extremely high inflation appears more common among lower-GDP-per-capita countries, while higher-GDP-per-capita countries tend to show more stable inflation.

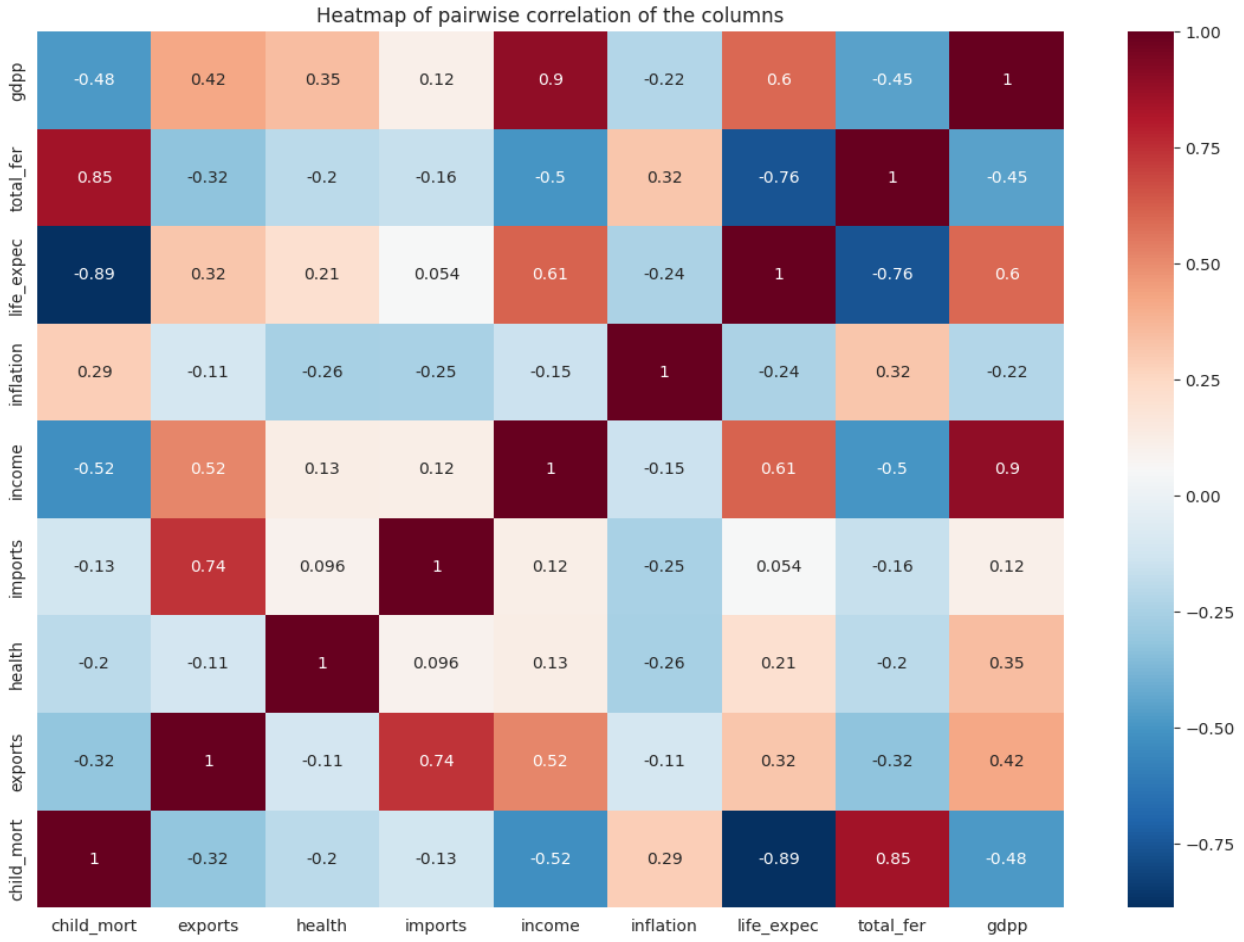


Figure 6: Heatmap of pairwise correlation of the columns

4.2.2 Correlation Heatmap

A Pearson correlation heatmap was constructed to quantify linear relationships and identify strongly co-related variables. Key observations include:

- **Strong negative correlations:** child_mort vs. life_expec (≈ -0.89), total_fer vs. life_expec (≈ -0.77), and child_mort vs. gdpp (≈ -0.83). These indicate that higher mortality and fertility tend to coincide with lower life expectancy and lower economic development.
- **Strong positive correlations:** income vs. gdpp (≈ 0.90), income vs. exports (≈ 0.73), and gdpp vs. exports (≈ 0.77), highlighting the interconnected nature of economic prosperity and trade activity.
- **Moderate correlations:** health shows meaningful associations with life expectancy and income, suggesting that greater health investment is linked with improved outcomes and higher prosperity.

The heatmap provides a compact summary of interdependencies and helps anticipate multicollinearity considerations for downstream modeling.

4.3 Visualization Quality and Reporting Standards

All visualizations were produced using appropriate chart types for the analytical objective:

- histograms and box plots for distributional analysis and outlier detection,
- scatter plots for pairwise relationship inspection,
- pair plots for multivariate overview,
- a correlation heatmap for summarizing linear dependencies.

Figures were formatted with clear titles, labeled axes (including units where applicable), readable legends where necessary, and consistent styling to support interpretability.

4.4 Insights and Domain-Based Interpretation

The EDA reveals clear development-linked patterns: economically stronger countries tend to exhibit lower child mortality and higher life expectancy, while countries with high fertility and high child mortality often align with lower GDP per capita. Numerous outliers—particularly in inflation, income, GDP per capita, and child mortality—highlight extreme country profiles that are important for identifying high-need cases and for designing robust preprocessing steps prior to clustering.

5 Statistical Analysis

This section presents hypothesis-driven statistical analyses to validate relationships among key socio-economic and health indicators. The objective is to provide quantitative evidence supporting the patterns observed during the exploratory data analysis (EDA), using appropriate statistical tests and clearly interpreting their outputs.

5.1 Hypothesis Formulation

To assess relationships between critical indicators, the following hypotheses were defined.

Hypothesis 1: Child Mortality vs. Life Expectancy

H_0: There is **no significant relationship** between child mortality and life expectancy.

H_1: There is a **significant negative relationship** between child mortality and life expectancy.

Hypothesis 2: Income vs. GDP per Capita

H_0: Income has **no significant relationship** with GDP per capita.

H_1: Income has a **significant positive relationship** with GDP per capita.

5.2 Choice of Statistical Tests and Justification

For both hypotheses, the **Pearson correlation test** was used to quantify the strength and direction of *linear* association between two continuous variables. This test is appropriate here because:

- the variables are continuous (e.g., `child_mort`, `life_expec`, `income`, `gdpp`),
- the goal is to assess *linear* relationships,
- Pearson correlation is commonly used and is reasonably robust to mild deviations from normality, especially with moderate sample sizes.

5.3 Results and Statistical Outputs

The Pearson correlation coefficient r and corresponding p-values were computed for each hypothesis.

Pearson Correlation Results

- **Child Mortality vs. Life Expectancy:**
  - Correlation coefficient (r): -0.887
  - p-value: 3.72×10^{-57}
- **Income vs. GDP per Capita:**
  - Correlation coefficient (r): 0.896
  - p-value: 6.40×10^{-60}

5.4 Interpretation: p-values and Confidence Intervals

A significance level of $\alpha = 0.05$ was used:

- If $p < 0.05$, reject H_0.
- If $p \geq 0.05$, fail to reject H_0.

Hypothesis 1 (Child Mortality vs. Life Expectancy)

Since $p = 3.72 \times 10^{-57} < 0.05$, we **reject** H_0. The correlation ($r = -0.887$) indicates a **strong negative** linear association: countries with higher child mortality tend to have substantially lower life expectancy.

Hypothesis 2 (Income vs. GDP per Capita)

Since $p = 6.40 \times 10^{-60} < 0.05$, we **reject** H_0. The correlation ($r = 0.896$) indicates a **strong positive** linear association: higher income levels are associated with higher GDP per capita.

Confidence intervals. In addition to p-values, reporting a confidence interval (CI) for r provides an uncertainty range for the estimated correlation (commonly a 95% CI, often computed via Fisher's z-transformation). The CI supports interpretation by showing plausible values of the true correlation in the population.

6 Machine Learning Model Implementation

This section describes the implementation of unsupervised machine learning models to cluster countries based on socio-economic and health indicators. The objective is to group countries with similar profiles in order to identify clusters that may correspond to higher developmental need.

6.1 Model Selection Rationale

Because the task involves discovering structure in the data without ground-truth labels, **unsupervised learning** (clustering) is appropriate. Two clustering approaches were used:

- **K-Means:** Selected for its computational efficiency, simplicity, and interpretability when partitioning data into a pre-specified number of clusters.
- **Hierarchical Clustering:** Used as a complementary method to reveal nested structure and validate whether the selected number of clusters is consistent with the data's hierarchical relationships.

6.2 Feature Selection and Engineering

- **Feature selection:** The `country` column is an identifier (categorical) and was excluded from modeling. All remaining numerical indicators were used as features: `child_mort`, `exports`, `health`, `imports`, `income`, `inflation`, `life_expec`, `total_fer`, `gdpp`.

- **Scaling / transformation:** Since K-Means and Ward-based hierarchical clustering rely on distance computations, all numerical features were standardized using **Z-score normalization** (mean = 0, standard deviation = 1). This prevents large-scale variables (e.g., `income`, `gdpp`) from dominating the distance metric.

6.3 Training Pipeline and Experimental Setup

Unlike supervised learning, clustering does not require target labels; therefore, standard *train/test splitting* and *class balancing* are not directly applicable. Instead, the pipeline focused on preprocessing consistency, hyperparameter selection, and internal validation.

- **Pipeline steps:** data cleaning → feature selection → standardization → clustering model fit → cluster assignment → cluster profiling and interpretation.
- **Validation strategy:** Internal validation metrics (e.g., inertia/WCSS and silhouette score) and stability checks (multiple initializations in K-Means) were used to support model selection.

6.4 Model Optimization and Selection of Number of Clusters

To select an appropriate number of clusters k for K-Means, the following criteria were used:

- **Elbow method (WCSS/Inertia):** WCSS was computed for a range of k values. The elbow point, where marginal improvement decreases substantially, indicates a reasonable k. The plot (Figure 4.1) suggests an elbow around $k = 4$.
- **Silhouette score:** The silhouette score was computed across candidate k values to measure compactness and separation. Results (Figure 4.2) indicate that $k = 4$ provides a good balance.
- **KneeLocator:** Knee detection was used as an automated support tool and also identified $k = 4$ as the elbow point.

Based on these criteria, $k = 4$ was selected for the final K-Means model.

6.5 Training Pipeline: K-Means Clustering

K-Means clustering was implemented using `KMeans` from `sklearn.cluster`.

- **Configuration:** `n_clusters=4`, `init='k-means++'`, `n_init=10`, `max_iter=300`, `random_state=42`.
- **Fitting and assignment:** The standardized feature matrix was clustered using `fit_predict`, producing a cluster label for each country. The predicted labels were appended to the original dataset for downstream analysis (cluster profiling, summary statistics, and interpretation).

6.6 Training Pipeline: Hierarchical Clustering

Hierarchical clustering was applied as an alternative perspective on the country groupings.

- **Linkage method:** Ward linkage was used to minimize within-cluster variance and encourage compact clusters.

- **Dendrogram analysis:** A dendrogram was generated to visualize the hierarchical merging process. Cutting the dendrogram at a level consistent with four groups produced a 4-cluster solution.
- **Cluster extraction:** Cluster labels were extracted (e.g., via `fcluster`) and attached to a dataset copy for comparison with K-Means labels.

7 Evaluation and Comparison of Results

This section evaluates the clustering models and compares their effectiveness for identifying high-need countries. Since clustering is an unsupervised task, evaluation is performed using internal validity metrics and visual inspection, followed by interpretation-driven cluster profiling and actionable recommendations.

7.1 Performance Evaluation using Appropriate Metrics

To assess cluster quality objectively, three internal evaluation metrics were computed for both K-Means and Hierarchical Clustering:

- **Silhouette Score:** Measures how well each point fits within its assigned cluster compared to neighboring clusters. Values range from -1 to 1, where higher values indicate better separation and cohesion.
- **Davies–Bouldin Index (DBI):** Measures the average similarity between each cluster and its most similar cluster. Lower values indicate more compact and better-separated clusters.
- **Calinski–Harabasz Index (CHI):** Ratio of between-cluster dispersion to within-cluster dispersion. Higher values indicate better-defined clustering structure.

K-Means Evaluation (4 Clusters)

- Silhouette Score: 0.296
- Davies–Bouldin Index: 1.053
- Calinski–Harabasz Index: 62.241

Hierarchical Clustering Evaluation (4 Clusters)

- Silhouette Score: 0.248
- Davies–Bouldin Index: 1.080
- Calinski–Harabasz Index: 48.263

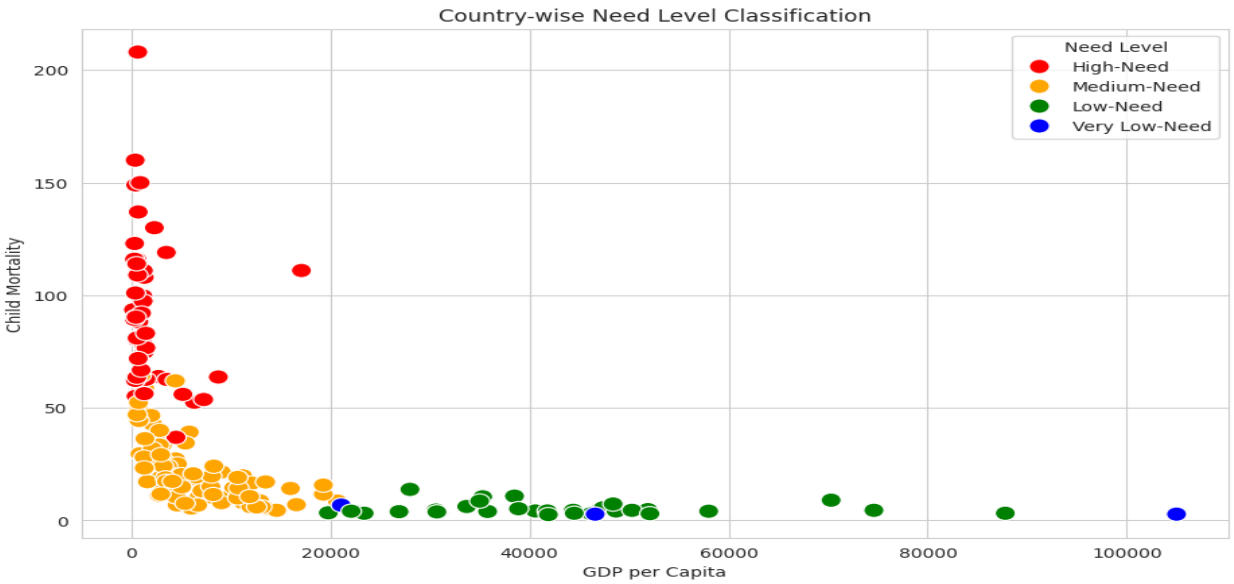


Figure 7: Child Mortality vs. GDP per Capita

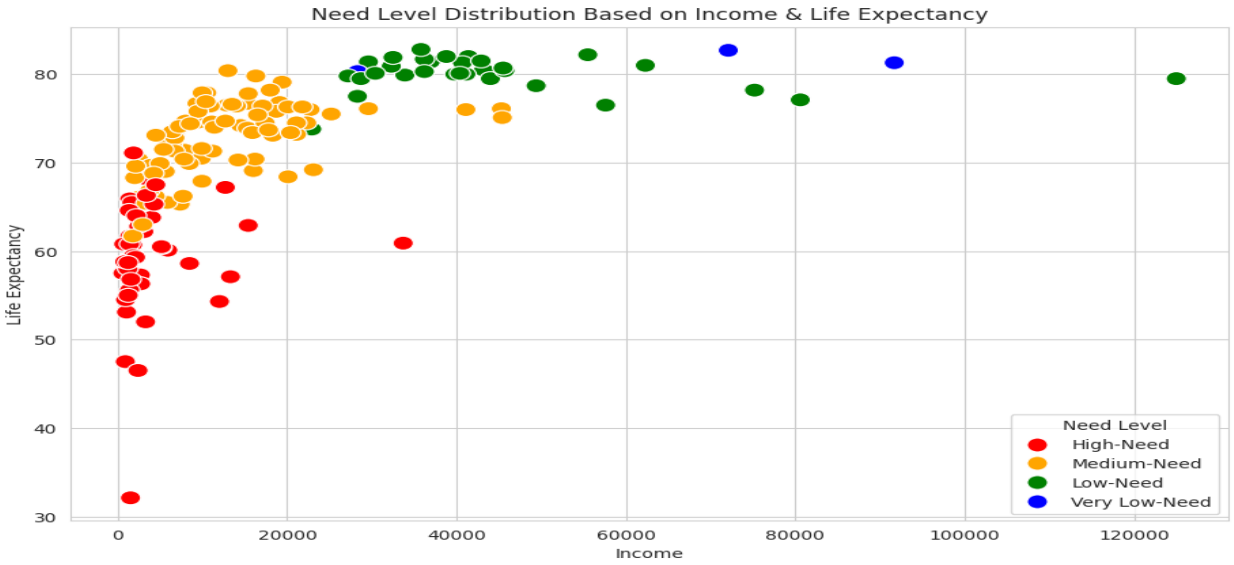


Figure 8: Life Expectancy vs. Income

7.2 Visual Evaluation: Cluster Distribution Plots

In addition to numeric metrics, scatter plots were used to visually assess cluster separation and interpretability. Countries were colored by assigned *Need Level* (derived from the K-Means cluster solution) across key development indicators:

- **Child Mortality vs. GDP per Capita (Figure 7):** The high-need cluster forms a distinct region characterized by high child mortality and low GDP per capita. Low-need clusters occupy low-mortality, high-GDP-per-capita regions, with medium-need countries in between.
- **Life Expectancy vs. Income (Figure 8):** High-need countries are concentrated at low income and low life expectancy, while low-need clusters show higher income and longer life expectancy.
- **Total Fertility vs. Health Spending (Figure 12):** High-need countries exhibit higher fertility rates and often lower health spending, while low-need clusters show lower fertility with comparatively higher health investment.
- **Inflation vs. GDP per Capita (Figure 11):** Separation is weaker than other plots, but high-need countries generally show lower GDP per capita with higher inflation volatility.

7.3 Comparison of Multiple Models with Reasoning

Table 2 summarizes the quantitative comparison between the two clustering methods.

Table 2: Comparison of K-Means and Hierarchical Clustering Performance

Metric	**K-Means (4 Clusters)**	**Hierarchical (4 Clusters)**
Silhouette Score	0.296	0.248
Davies–Bouldin Index	1.053	1.080
Calinski–Harabasz Index	62.241	48.263

Across all three metrics, **K-Means performs better**:

- higher Silhouette and CHI indicate improved cohesion and separation,
- lower DBI indicates reduced overlap between clusters.

Therefore, K-Means with 4 clusters was selected as the preferred clustering solution for downstream interpretation.

High-Need Cluster Profile (K-Means)

The high-need cluster (Cluster 1 in the notebook) is characterized by:

- high child mortality (mean: 92.96),
- low income (mean: 3942.40),
- low GDP per capita (mean: 1922.38),
- high total fertility (mean: 5.01),
- low life expectancy (mean: 59.19),
- relatively high inflation (mean: 12.02).

This profile is coherent and aligns with the study objective: identifying countries facing substantial health and economic challenges.

8 Conclusion

This project applied a complete data-mining workflow to identify countries in critical need of aid using socio-economic and health indicators. After data quality assessment and standard preprocessing, exploratory analysis revealed significant global inequality, with right-skewed distributions and notable outliers across several variables. Correlation analysis aligned with development theory, showing strong positive relationships among prosperity indicators (e.g., income and GDP per capita) and strong negative relationships between adverse conditions (e.g., child mortality, fertility) and life expectancy.

Statistical analysis confirmed these patterns, including a significant negative correlation between child mortality and life expectancy and a strong positive correlation between income and GDP per capita. For unsupervised segmentation, K-Means and Hierarchical clustering were evaluated using internal validation metrics. K-Means with $k = 4$ outperformed Hierarchical clustering and produced interpretable clusters representing *High-Need*, *Medium-Need*, *Low-Need*, and *Very Low-Need* countries. The High-Need cluster was characterized by high child mortality and fertility, low income and GDP per capita, and low life expectancy, making it suitable for prioritizing development interventions.

This study highlights the importance of preprocessing, exploratory analysis, and metric-based validation in clustering tasks. Limitations include the use of a static dataset, a limited feature set, and the absence of external ground-truth labels. Future work may incorporate temporal data, additional development indicators, alternative clustering methods, and dynamic frameworks for continuously updating country need levels.

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A Appendices

A.1 Complete Source Code

This appendix provides the full Python implementation for the "High-Need Country" identification project, including data cleaning, exploratory visualizations, statistical testing, and clustering models.

Listing 1: Data Loading and Preprocessing

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.pyplot import figure
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from scipy.stats import pearsonr
import statsmodels.api as sm
from sklearn.cluster import KMeans
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster
from sklearn.cluster import AgglomerativeClustering
from sklearn.metrics import silhouette_score, davies_bouldin_score, calinski_harabasz_score

# 1. Data Loading and Quality Check
df = pd.read_csv('Country-data.csv')

# 2. Data Transformation
country_names = df['country']
df_numeric = df.drop(columns=['country'])

# Z-score Standardization
standardized_df = df_numeric.copy()
for col in df_numeric.columns:
    standardized_df[col] = (df_numeric[col] - df_numeric[col].mean()) / df_numeric[col].std
        ()
```

Listing 2: Exploratory Data Analysis (EDA)

```
# Univariate Analysis: Histograms and Boxplots
numerical_columns = ['child_mort', 'exports', 'health', 'imports', 'income', 'inflation', '
    life_expec', 'total_fer', 'gdpp']

# Histograms
for col in numerical_columns:
    plt.figure(figsize=(16, 6))
    plt.hist(df_numeric[col], bins=20, facecolor='b')
    plt.title(f'Histogram of {col}')
    plt.show()

# Boxplots for Outlier Detection
for col in numerical_columns:
    plt.figure(figsize=(6, 7))
    plt.boxplot(df_numeric[col])
    plt.title(f'Boxplot for {col}')
    plt.show()

# Bivariate Analysis: Heatmaps and Scatterplots
f, ax = plt.subplots(figsize=(14, 10))
corr = df_numeric.corr()
sns.heatmap(corr, cmap="RdBu_r", annot=True)
plt.title("Heatmap of Pairwise Correlation")
plt.show()
```

Listing 3: Statistical Analysis and Hypothesis Testing

```
# Hypothesis 1: child_mort vs life_expec
corr1, pval1 = pearsonr(df_numeric['child_mort'], df_numeric['life_expec'])

# Hypothesis 2: income vs gdpp
corr2, pval2 = pearsonr(df_numeric['income'], df_numeric['gdpp'])

# Q-Q Plots for Normality Check
for col in ['child_mort', 'income', 'gdpp', 'life_expec']:
    sm.qqplot(df_numeric[col], line='s')
    plt.title(f'Q-Q Plot for {col}')
    plt.show()
```

Listing 4: Clustering Implementation (K-Means and Hierarchical)

```
# Elbow Method to find K
wcss = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, init='k-means++', n_init=10, random_state=42)
    kmeans.fit(standardized_df)
    wcss.append(kmeans.inertia_)

# Final K-Means Model
kmeans_final = KMeans(n_clusters=4, random_state=42, n_init=10)
kmeans_labels = kmeans_final.fit_predict(standardized_df)
df_kmeans = df.copy()
df_kmeans['cluster'] = kmeans_labels
kmeans_mapping = {1: "High-Need", 2: "Medium-Need", 0: "Low-Need", 3: "Very Low-Need"}
df_kmeans['Need_Level'] = df_kmeans['cluster'].map(kmeans_mapping)

# Hierarchical Clustering
linked = linkage(standardized_df, method='ward')
hc_labels = fcluster(linked, t=4, criterion='maxclust')
df_hc = df.copy()
df_hc['hc_cluster'] = hc_labels
```

Listing 5: Evaluation and Visualizations

```
# Evaluation Scores
print(f"K-Means Silhouette Score: {silhouette_score(standardized_df, kmeans_labels):.3f}")
print(f"Hierarchical Silhouette Score: {silhouette_score(standardized_df, hc_labels):.3f}")

# Final Scatterplots of Need Levels
plt.figure(figsize=(11,7))
sns.scatterplot(data=df_kmeans, x='gdpp', y='child_mort', hue='Need_Level',
                palette={"High-Need": "red", "Medium-Need": "orange", "Low-Need": "green", "
                    Very Low-Need": "blue"}, s=120)
plt.title("Country-wise Need Level Classification")
plt.show()
```

A.2 Additional visualizations

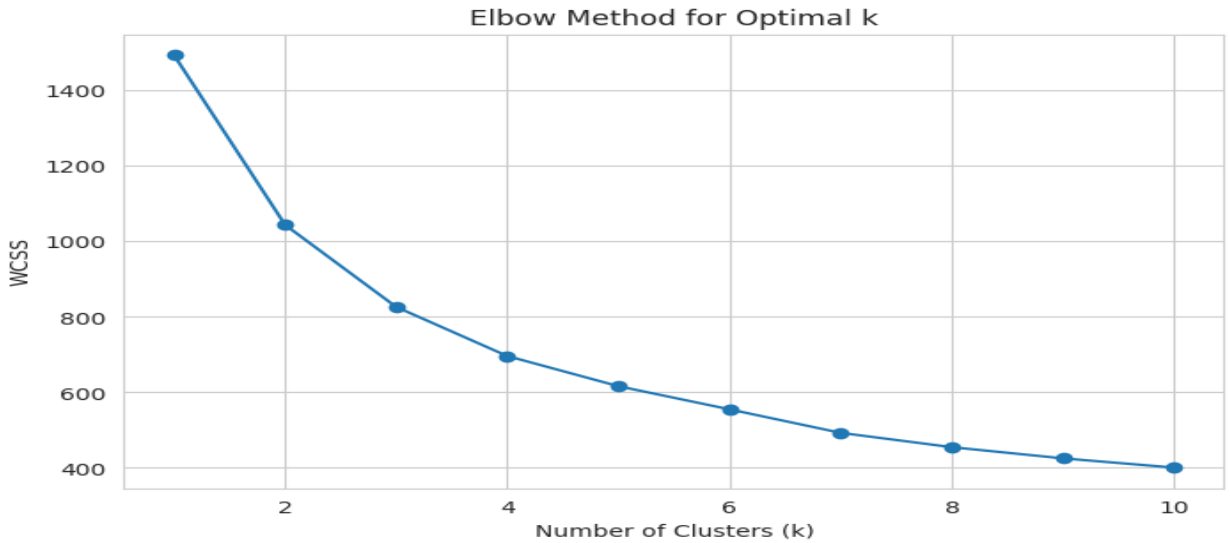


Figure 9: Elbow Curve

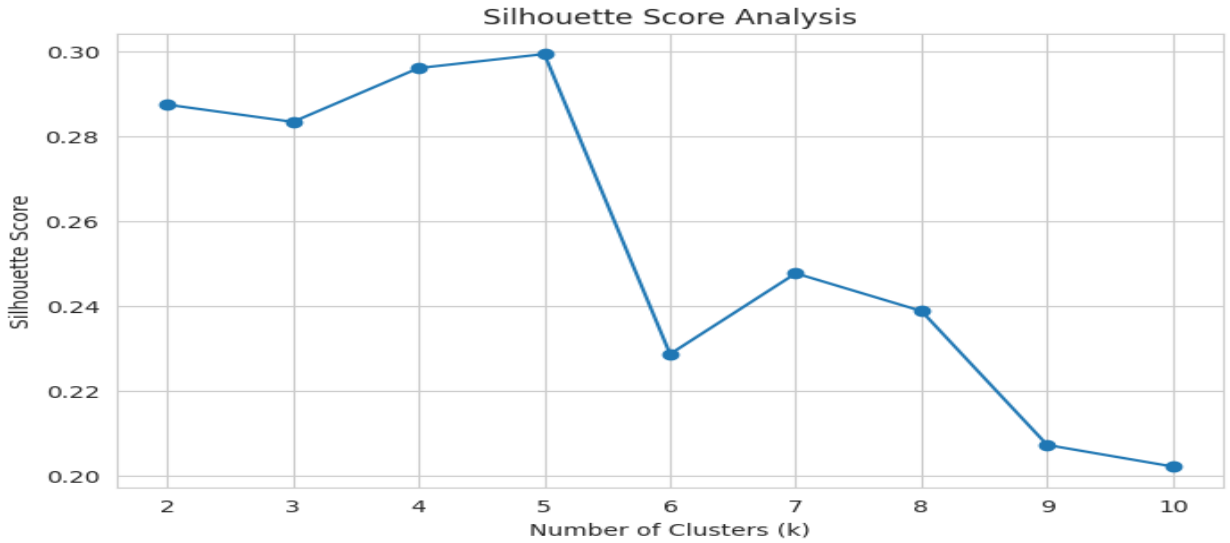


Figure 10: Silhouette Score

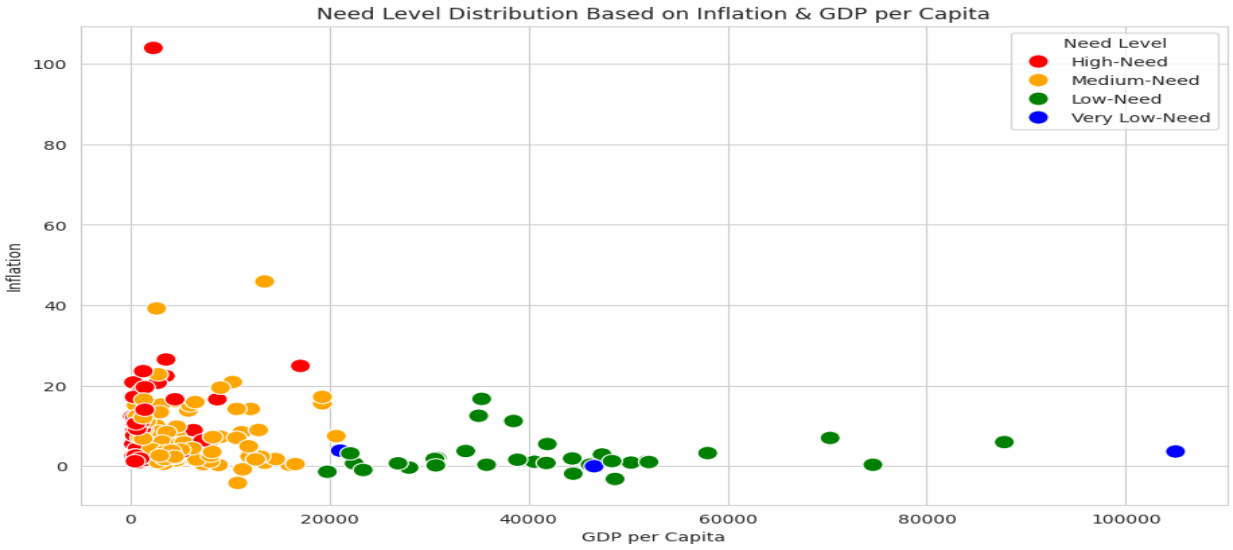


Figure 11: Inflation vs. GDP per Capita

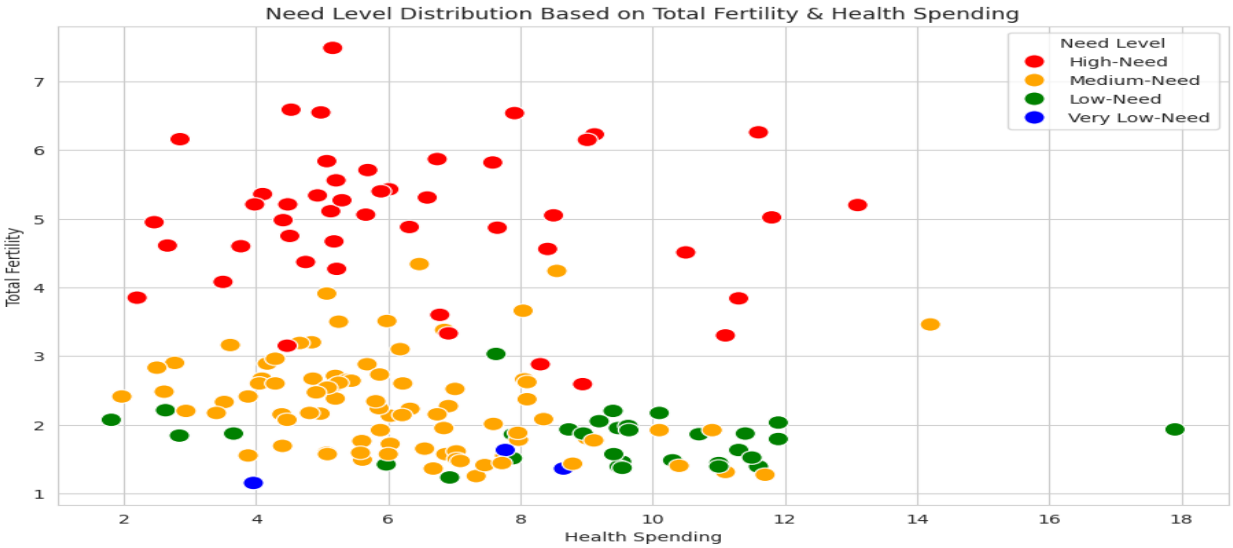


Figure 12: Total Fertility vs. Health Spending

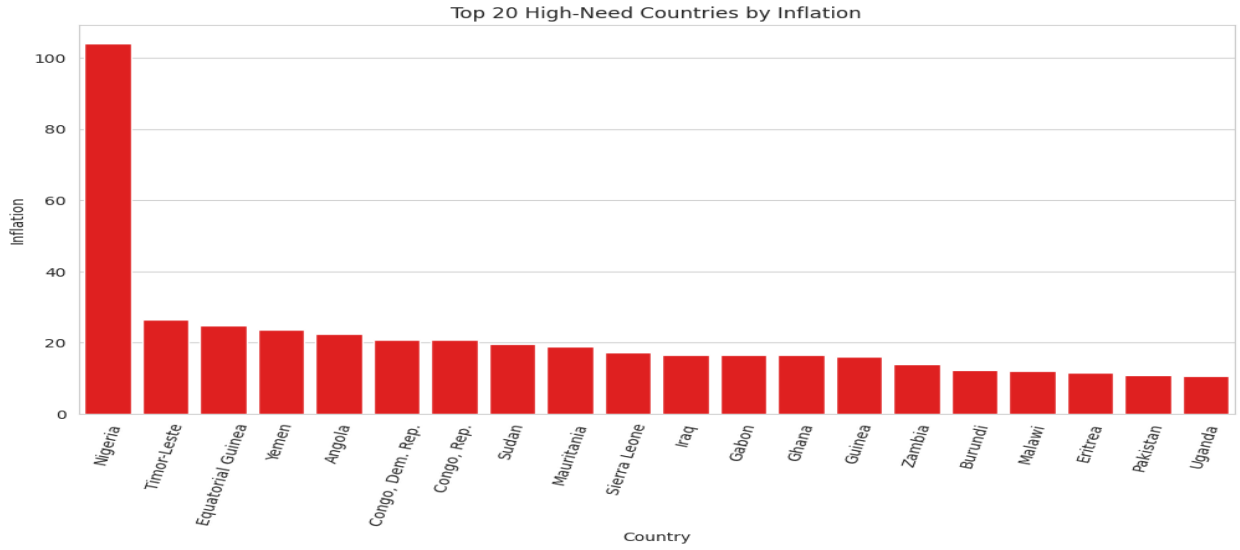


Figure 13: Top 20 High-Need Countries by Inflation

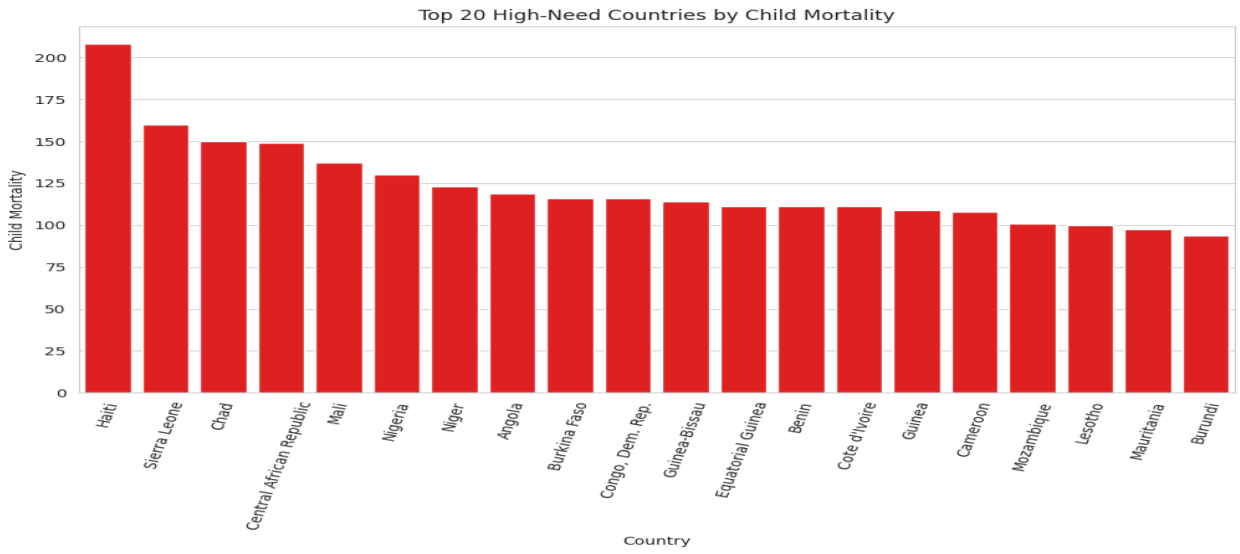


Figure 14: Top 20 High-Need Countries by Child Mortality